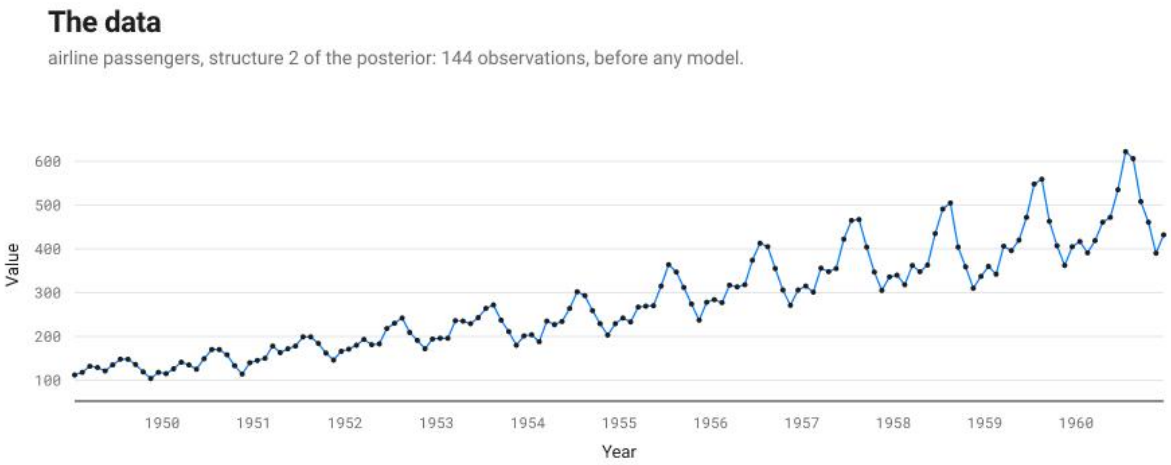
An automatic report for the dataset: airline passengers, structure 2 of the posterior

`autostat` (automatic-statistician-revisited)

08 November 2026

Produced by ***`autostat`****, a Python 3 port of the Automatic Bayesian Covariance Discovery system of Lloyd et al. (AAAI 2014).*



The data

Executive summary

With posterior weight 0.276, AutoGP kept `WN + (C + Lin + Lin + Per + GE x (C + Lin)) x (C + Lin) x (C + Lin)` (Bayesian information criterion 1207.1; negative log likelihood 558.8 on 144 points). It decomposes into 25 additive components, listed by how much each reduces extrapolation error:

1. A linearly increasing function.
2. A quadratic polynomial.
3. A linearly increasing function.
4. A quadratic polynomial.
5. A linearly increasing function.
6. A quadratic polynomial.
7. A rough function with marginal standard deviation increasing linearly away from Dec 1950.

8. A cubic polynomial.
9. A quadratic polynomial.
10. A quadratic polynomial.
11. A rough function with marginal standard deviation increasing linearly away from Nov 1951.
12. A periodic function with a period of 1.0 years.
13. A rough function with quadraticly varying marginal standard deviation.
14. A periodic function with a period of 1.0 years but with quadraticly varying amplitude.
15. A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Dec 1950.
16. A rough function with quadraticly varying marginal standard deviation.
17. A rough function with marginal standard deviation increasing linearly away from Aug 1953.
18. A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Nov 1951.
19. A rough function with quadraticly varying marginal standard deviation.
20. Uncorrelated noise.
21. A rough function with cubicly varying marginal standard deviation.
22. A cubic polynomial.
23. A rough function.
24. A linearly increasing function.
25. A constant.

#	R^2 cumulative (%)	Residual variance explained (%)	CV MAE	MAE reduction (%)
data			280	
1	49.3	49.3	234	16.5
2	69.0	38.7	213	9.1
3	81.1	39.2	193	9.5
4	82.4	6.7	190	1.6
5	85.4	17.1	187	1.4
6	85.5	1.0	186	0.7
7	85.9	2.2	185	0.5
8	85.9	0.6	184	0.5
9	86.0	0.6	183	0.4
10	86.1	0.6	182	0.4
11	86.2	0.7	182	0.2
12	93.0	49.3	182	0.2
13	93.1	0.8	181	0.2
14	96.8	53.9	181	0.1
15	99.2	76.3	181	0.1
16	99.3	4.8	181	0.1
17	99.3	9.6	181	0.1
18	99.5	29.1	180	0.1
19	99.5	2.4	180	0.0
20	99.7	37.6	180	0.0
21	99.7	1.7	181	-0.0
22	99.7	9.2	181	-0.2
23	99.9	67.9	182	-0.6
24	100.0	100.0	192	-5.5
25	100.0	-4.7	281	-46.5

MAE: mean absolute error over ten contiguous held-out blocks; CV: cross-validated.

The formula found

The structure search is a symbolic regression: it searches a grammar of expressions over kernels and keeps the one with the lowest Bayesian information criterion. This is the expression it kept, with every fitted parameter in place; t is in years.

$$\begin{aligned} y(t) = 280.3 &+ f_1(t) + f_2(t) + f_3(t) + f_4(t) + f_5(t) + f_6(t) + f_7(t) + f_8(t) + f_9(t) \\ &+ f_{10}(t) + f_{11}(t) + f_{12}(t) + f_{13}(t) + f_{14}(t) + f_{15}(t) + f_{16}(t) + f_{17}(t) \\ &+ f_{18}(t) + f_{19}(t) + f_{20}(t) + f_{21}(t) + f_{22}(t) + f_{23}(t) + f_{24}(t) + f_{25}(t) \end{aligned}$$

Each component is a Gaussian process with its own covariance:

$$f_1(t) \sim \mathrm{GP}\big(0,\ \ 62.76\,(t-1951)(t'-1951)\big)$$

$$f_2(t) \sim \mathrm{GP}\big(0,\ \ 0.8172\,(t-1951)(t'-1951)\cdot(t-1952)(t'-1952)\big)$$

$$f_3(t) \sim \mathrm{GP}\big(0,\ \ 30.82\,(t-1952)(t'-1952)\big)$$

$$f_4(t) \sim \mathrm{GP}\big(0,\ \ 0.3222\,(t-1954)(t'-1954)\cdot(t-1951)(t'-1951)\big)$$

$$f_5(t) \sim \mathrm{GP}\big(0,\ \ 12.15\,(t-1954)(t'-1954)\big)$$

$$f_6(t) \sim \mathrm{GP}\big(0,\ 0.03642\,(t-1965)(t'-1965)\cdot(t-1951)(t'-1951)\big)$$

$$f_7(t) \sim \mathrm{GP}\left(0,\ 1.937\,\exp\left(-\left(\frac{|t-t'|}{5.643}\right)^{0.6898}\right)\cdot(t-1951)(t'-1951)\right)$$

$$f_8(t) \sim \mathrm{GP}\Big(0,\ 4.742\times10^{-4}\,(t-1951)(t'-1951)\cdot(t-1965)(t'-1965)\cdot(t-1952)(t'-1952)\Big)$$

$$f_9(t) \sim \mathrm{GP}\big(0,\ 0.1582\,(t-1954)(t'-1954)\cdot(t-1952)(t'-1952)\big)$$

$$f_{10}(t) \sim \mathrm{GP}\big(0,\ 0.01789\,(t-1965)(t'-1965)\cdot(t-1952)(t'-1952)\big)$$

$$f_{11}(t) \sim \mathrm{GP}\left(0,\ 0.951\,\exp\left(-\left(\frac{|t-t'|}{5.643}\right)^{0.6898}\right)\cdot(t-1952)(t'-1952)\right)$$

$$f_{12}(t) \sim \mathrm{GP}\left(0,\ 660.9\,\frac{\exp\Big(4.213\cos\frac{2\pi(t-t')}{0.9981}\Big)-I_0(4.213)}{e^{4.213}-I_0(4.213)}\right)$$

$$f_{13}(t) \sim \mathrm{GP}\left(0,\ 0.02522\,\exp\left(-\left(\frac{|t-t'|}{5.643}\right)^{0.6898}\right)\cdot(t-1951)(t'-1951)\cdot(t-1952)(t'-1952)\right)$$

$$f_{14}(t) \sim \mathrm{GP}\left(0,\ 0.2282\,(t-1951)(t'-1951)\cdot(t-1952)(t'-1952)\cdot\frac{\exp\Big(4.213\cos\frac{2\pi(t-t')}{0.9981}\Big)-I_0(4.213)}{e^{4.213}-I_0(4.213)}\right)$$

$$f_{15}(t) \sim \mathrm{GP}\left(0,\ 17.53\,(t-1951)(t'-1951)\cdot\frac{\exp\Big(4.213\cos\frac{2\pi(t-t')}{0.9981}\Big)-I_0(4.213)}{e^{4.213}-I_0(4.213)}\right)$$

$$f_{16}(t) \sim \mathrm{GP}\left(0,\ 0.03018\,\exp\left(-\left(\frac{|t-t'|}{5.643}\right)^{0.6898}\right)\cdot(t-1951)(t'-1951)\cdot(t-1954)(t'-1954)\right)$$

$$f_{17}(t) \sim \mathrm{GP}\left(0,\ 1.138\,\exp\left(-\left(\frac{|t-t'|}{5.643}\right)^{0.6898}\right)\cdot(t-1954)(t'-1954)\right)$$

$$f_{18}(t) \sim \mathrm{GP}\left(0,\ 8.605\,(t-1952)(t'-1952)\cdot\frac{\exp\Big(4.213\cos\frac{2\pi(t-t')}{0.9981}\Big)-I_0(4.213)}{e^{4.213}-I_0(4.213)}\right)$$

$$f_{19}(t) \sim \mathrm{GP}\left(0,\ 0.01482\,\exp\left(-\left(\frac{|t-t'|}{5.643}\right)^{0.6898}\right)\cdot(t-1954)(t'-1954)\cdot(t-1952)(t'-1952)\right)$$

$$f_{20}(t) \sim \mathrm{GP}(0,\ \ 32.15\,\delta_{tt'})$$

$$f_{21}(t) \sim \mathrm{GP}\Bigg(0,\ \ 3.93 \times 10^{-4}\,\exp\Bigg(-\left(\frac{|t-t'|}{5.643}\right)^{0.6898}\Bigg)\cdot(t-1951)(t'-1951)\cdot(t-1954)(t'-1954)\cdot(t-1952)(t'-1952)\Bigg)$$

$$f_{22}(t) \sim \mathrm{GP}\big(0,\ \ 0.004195\,(t-1951)(t'-1951)\cdot(t-1954)(t'-1954)\cdot(t-1952)(t'-1952)\big)$$

$$f_{23}(t) \sim \mathrm{GP}\left(0,\ \ 73.04\,\exp\left(-\left(\frac{|t-t'|}{5.643}\right)^{0.6898}\right)\right)$$

$$f_{24}(t) \sim \mathrm{GP}\big(0,\ \ 1.374\,(t-1965)(t'-1965)\big)$$

$$f_{25}(t) \sim \mathrm{GP}(0,\ \ 2367)$$

Polynomial and periodic components have a closed form, read from their posterior mean:

$$f_1(t) = -21.47 + 11.17\,(t-1949)$$

$$f_2(t) = 4.821 - 4.206\,(t-1949) + 0.8831\,(t-1949)^2$$

$$f_3(t) = -18.64 + 6.559\,(t-1949)$$

$$f_4(t) = 2.125 - 1.514\,(t-1949) + 0.2124\,(t-1949)^2$$

$$f_5(t) = -19.13 + 3.672\,(t-1949)$$

$$f_6(t) = -1.449 + 0.8439\,(t-1949) - 0.04677\,(t-1949)^2$$

$$f_8(t) = 0.4651 - 0.4346\,(t-1949) + 0.1103\,(t-1949)^2 - 0.005283\,(t-1949)^3$$

$$f_9(t) = 0.892 - 0.4851\,(t-1949) + 0.06024\,(t-1949)^2$$

$$f_{10}(t) = -1.592 + 0.6587\,(t-1949) - 0.03473\,(t-1949)^2$$

$$\begin{aligned} f_{12}(t) \approx\ & 19.82\cos\left(\frac{2\pi\cdot(t-1949)}{0.9981} - 3.112\right) + 12.81\cos\left(\frac{2\pi\cdot 2\cdot(t-1949)}{0.9981} - 1.541\right) \\ & + 4.908\cos\left(\frac{2\pi\cdot 3\cdot(t-1949)}{0.9981} + 1.252\right) + 5.664\cos\left(\frac{2\pi\cdot 4\cdot(t-1949)}{0.9981} + 0.2723\right) \\ & + 2.459\cos\left(\frac{2\pi\cdot 5\cdot(t-1949)}{0.9981} - 0.08673\right) + 0.3101\cos\left(\frac{2\pi\cdot 6\cdot(t-1949)}{0.9981} + 1.735\right) \\ & + 0.437\cos\left(\frac{2\pi\cdot 7\cdot(t-1949)}{0.9981} + 2.349\right) + 0.1038\cos\left(\frac{2\pi\cdot 8\cdot(t-1949)}{0.9981} + 1.763\right) \end{aligned}$$

$$f_{22}(t) = -0.07817 + 0.0832\,(t-1949) - 0.02741\,(t-1949)^2 + 0.002748\,(t-1949)^3$$

$$f_{24}(t) = -15.82 + 0.9813\,(t-1949)$$

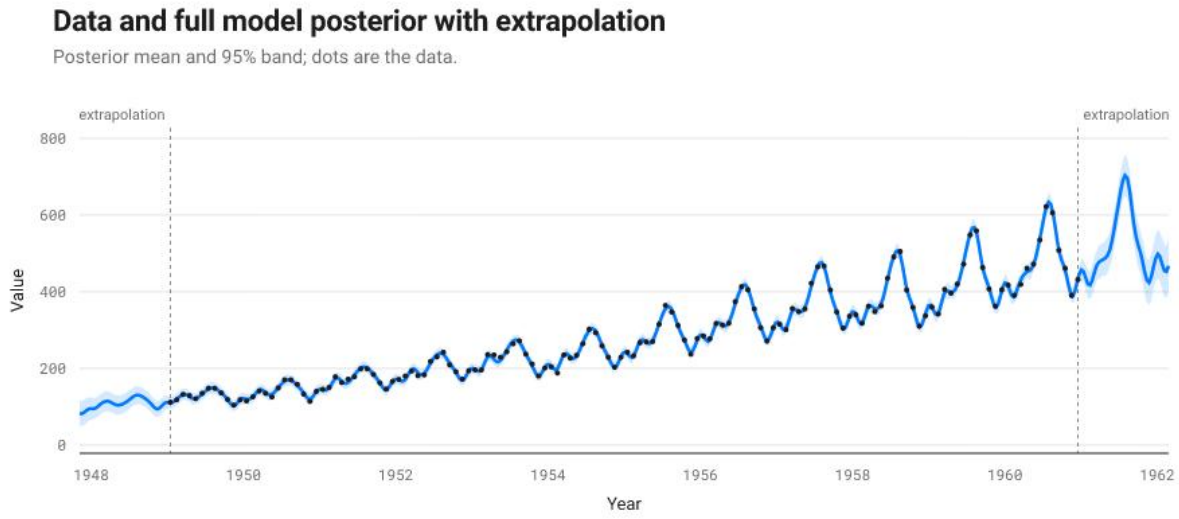
$$f_{25}(t) = -89.38$$

A periodic kernel is exactly a sum of cosines, so a periodic component is a Fourier series in its period; it is written up to the harmonic after which the rest weighs less than 0.1

The other components are smooth, quasi-periodic or noise: a draw from such a process has no finite closed form; its covariance above is the formula.

Closed form of each component:

#	component	closed form
1	A linearly increasing function	exact polynomial of degree 1
2	A quadratic polynomial	exact polynomial of degree 2
3	A linearly increasing function	exact polynomial of degree 1
4	A quadratic polynomial	exact polynomial of degree 2
5	A linearly increasing function	exact polynomial of degree 1
6	A quadratic polynomial	exact polynomial of degree 2
7	A rough function with marginal standard deviation increasing linearly away from Dec 1950	none: the covariance is the formula
8	A cubic polynomial	exact polynomial of degree 3
9	A quadratic polynomial	exact polynomial of degree 2
10	A quadratic polynomial	exact polynomial of degree 2
11	A rough function with marginal standard deviation increasing linearly away from Nov 1951	none: the covariance is the formula
12	A periodic function with a period of 1.0 years	Fourier series, 8 harmonics, largest gap 0.017 % of the component
13	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
14	A periodic function with a period of 1.0 years but with quadraticly varying amplitude	none: the covariance is the formula
15	A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Dec 1950	none: the covariance is the formula
16	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
17	A rough function with marginal standard deviation increasing linearly away from Aug 1953	none: the covariance is the formula
18	A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Nov 1951	none: the covariance is the formula
19	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
20	Uncorrelated noise	none: the covariance is the formula
21	A rough function with cubicly varying marginal standard deviation	none: the covariance is the formula
22	A cubic polynomial	exact polynomial of degree 3
23	A rough function	none: the covariance is the formula
24	A linearly increasing function	exact polynomial of degree 1
25	A constant	exact polynomial of degree 0



Data and full posterior

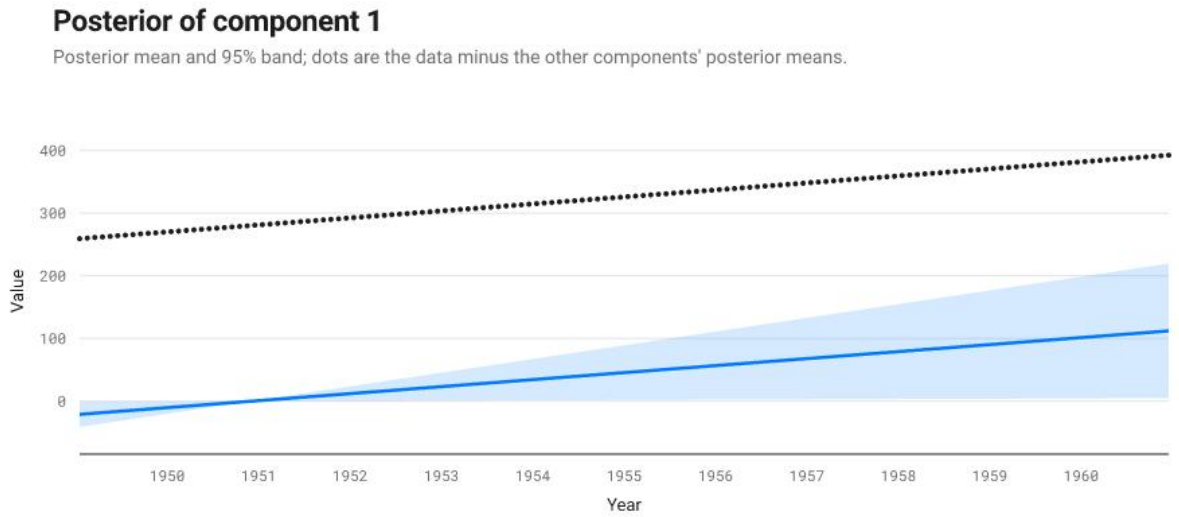
Detailed discussion of the components

Component 1: A linearly increasing function

This component is linearly increasing.

This component explains 49.3% of the data's variance on its own; with the components above it, 49.3%. Adding it changes the cross-validated MAE by -16.5%.

Extrapolation. This component is assumed to continue to increase linearly.



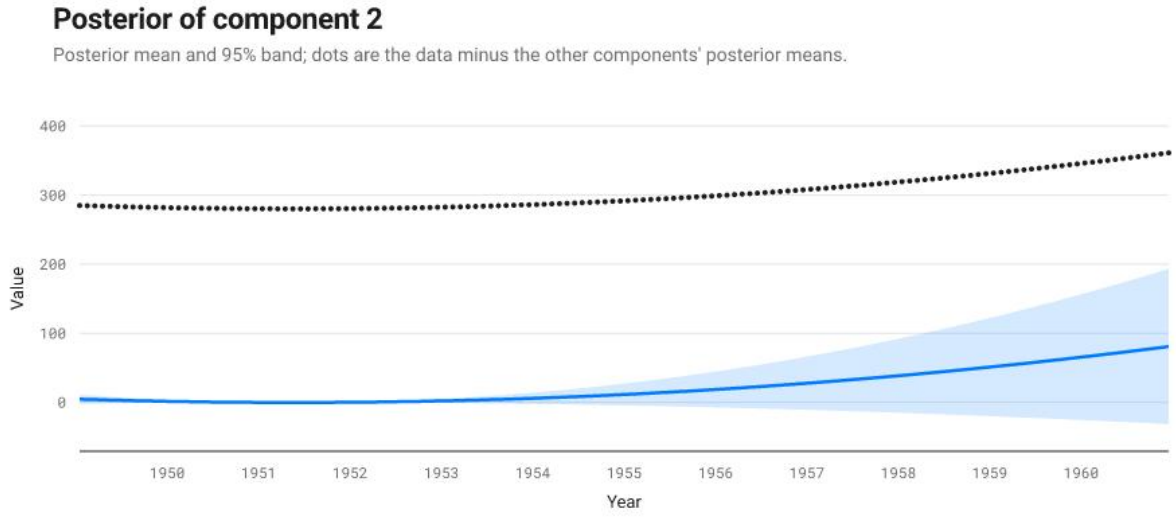
Component 1

Component 2: A quadratic polynomial

This component is a quadratic polynomial.

This component explains 31.6% of the data's variance on its own; with the components above it, 69.0%. Adding it changes the cross-validated MAE by -9.1%.

Extrapolation. This component is assumed to continue as a quadratic polynomial.



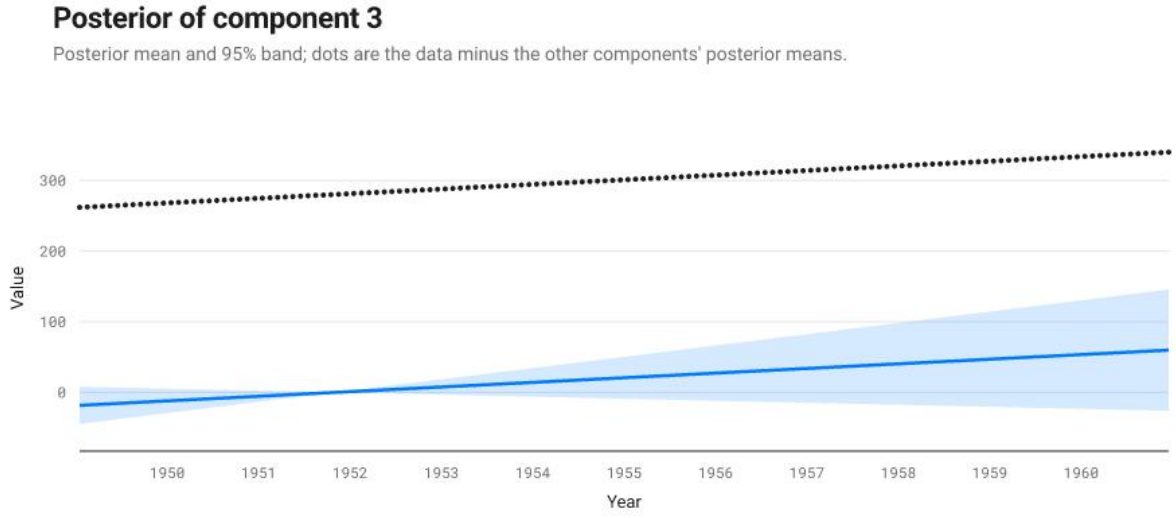
Component 2

Component 3: A linearly increasing function

This component is linearly increasing.

This component explains 31.5% of the data's variance on its own; with the components above it, 81.1%. Adding it changes the cross-validated MAE by -9.5%.

Extrapolation. This component is assumed to continue to increase linearly.



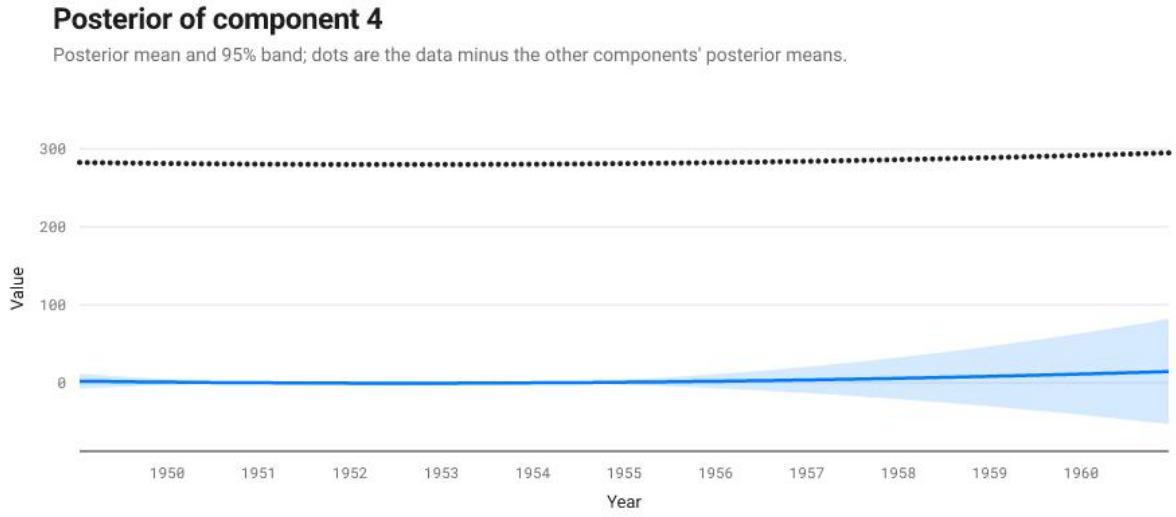
Component 3

Component 4: A quadratic polynomial

This component is a quadratic polynomial.

This component explains 5.8% of the data's variance on its own; with the components above it, 82.4%. Adding it changes the cross-validated MAE by -1.6%.

Extrapolation. This component is assumed to continue as a quadratic polynomial.



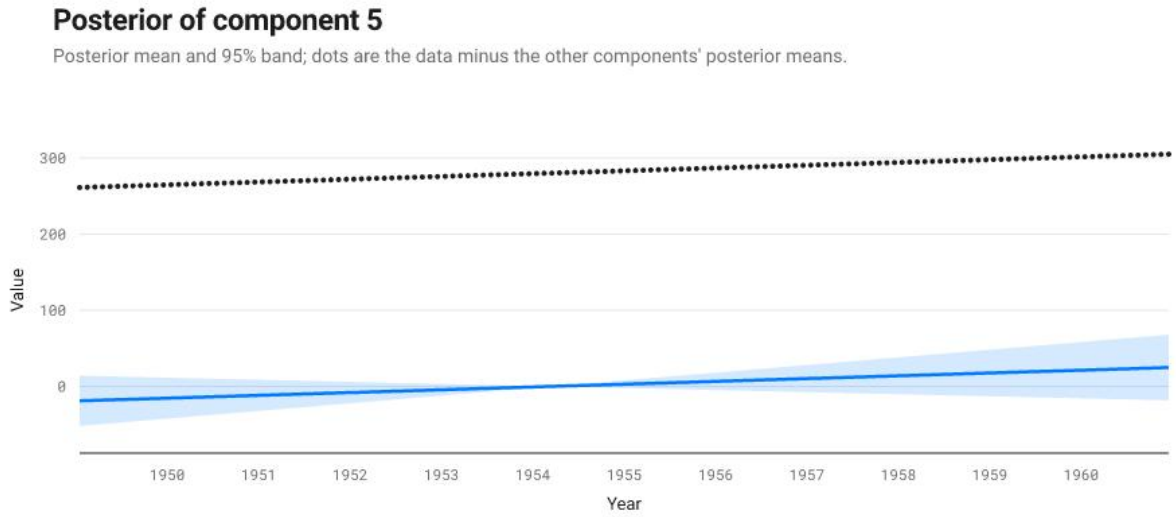
Component 4

Component 5: A linearly increasing function

This component is linearly increasing.

This component explains 18.5% of the data's variance on its own; with the components above it, 85.4%. Adding it changes the cross-validated MAE by -1.4%.

Extrapolation. This component is assumed to continue to increase linearly.



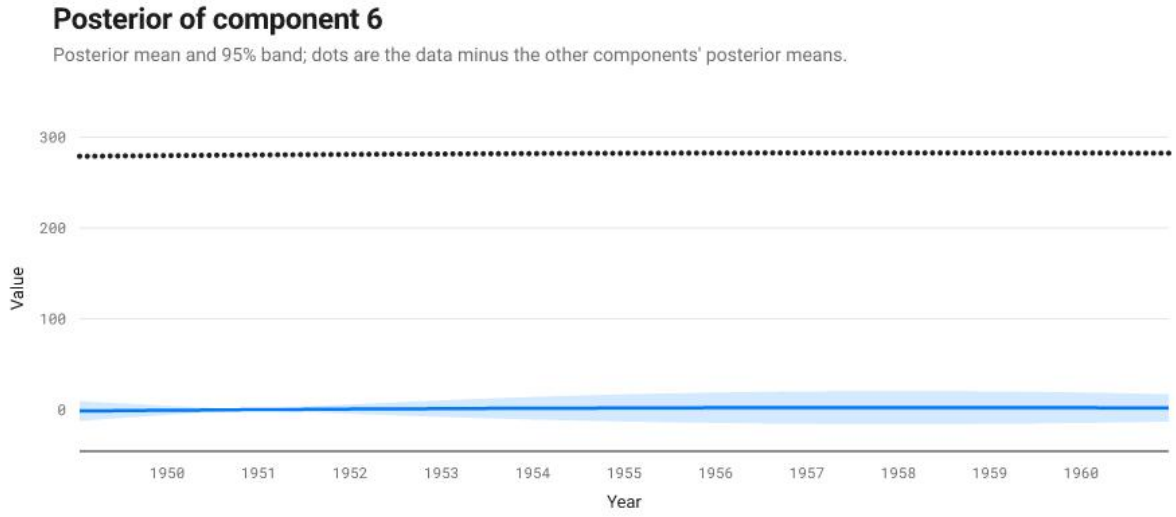
Component 5

Component 6: A quadratic polynomial

This component is a quadratic polynomial.

This component explains 1.4% of the data's variance on its own; with the components above it, 85.5%. Adding it changes the cross-validated MAE by -0.7%.

Extrapolation. This component is assumed to continue as a quadratic polynomial.



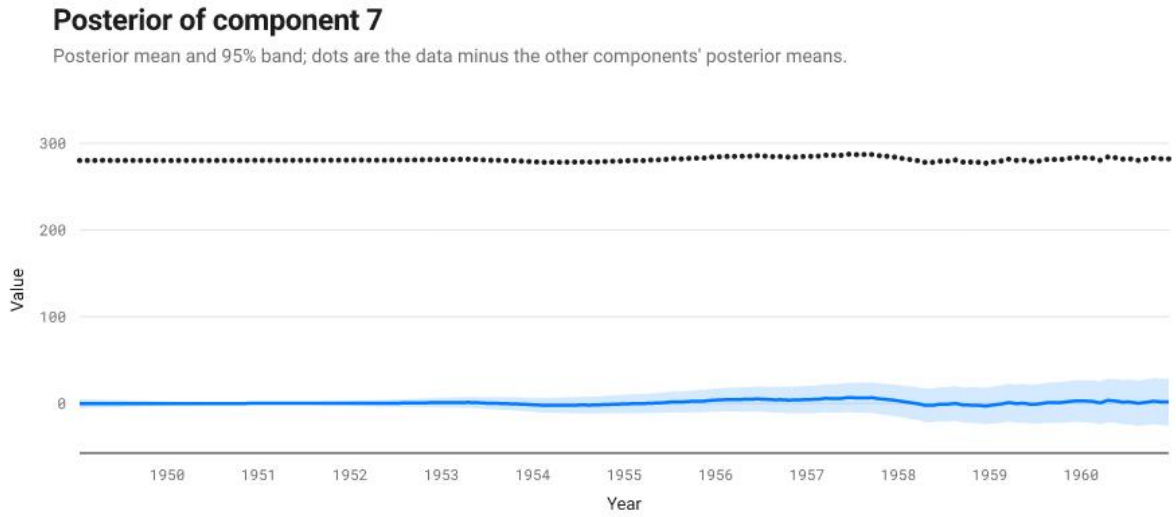
Component 6

Component 7: A rough function with marginal standard deviation increasing linearly away from Dec 1950

This component is a rough function with a typical lengthscale of 2.1 years. The marginal standard deviation of the function increases linearly away from Dec 1950. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.69).

This component explains 1.3% of the data's variance on its own; with the components above it, 85.9%. Adding it changes the cross-validated MAE by -0.5%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 2.1 years. The marginal standard deviation of the function is assumed to continue to increase linearly.



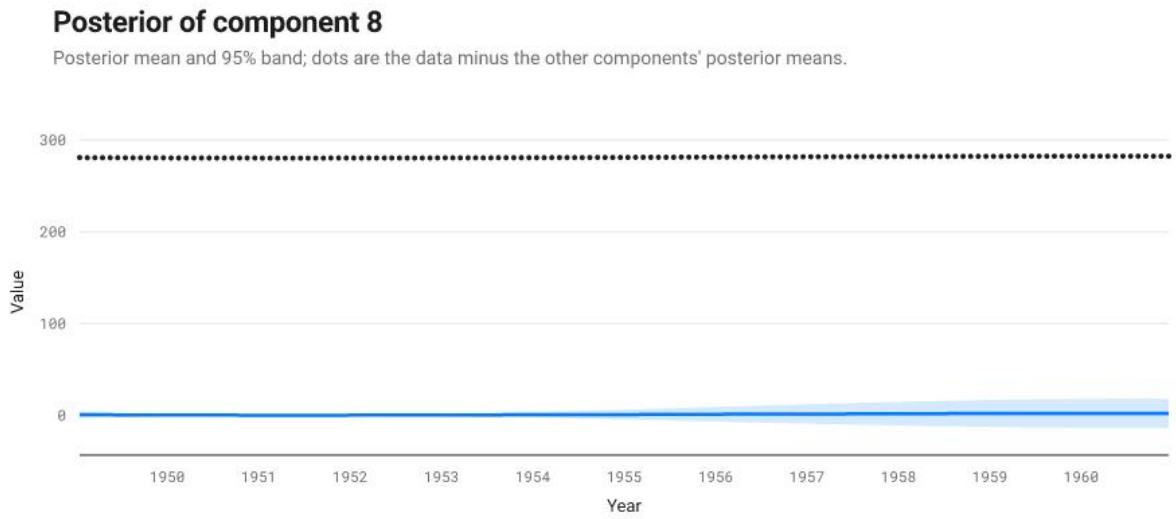
Component 7

Component 8: A cubic polynomial

This component is a cubic polynomial.

This component explains 1.1% of the data's variance on its own; with the components above it, 85.9%. Adding it changes the cross-validated MAE by -0.5%.

Extrapolation. This component is assumed to continue as a cubic polynomial.



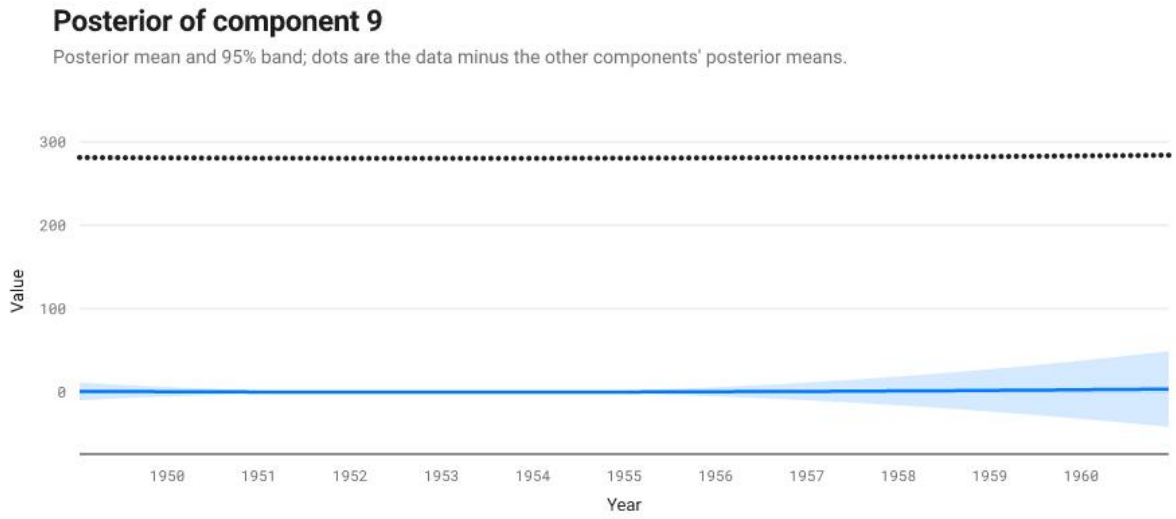
Component 8

Component 9: A quadratic polynomial

This component is a quadratic polynomial.

This component explains 1.4% of the data's variance on its own; with the components above it, 86.0%. Adding it changes the cross-validated MAE by -0.4%.

Extrapolation. This component is assumed to continue as a quadratic polynomial.



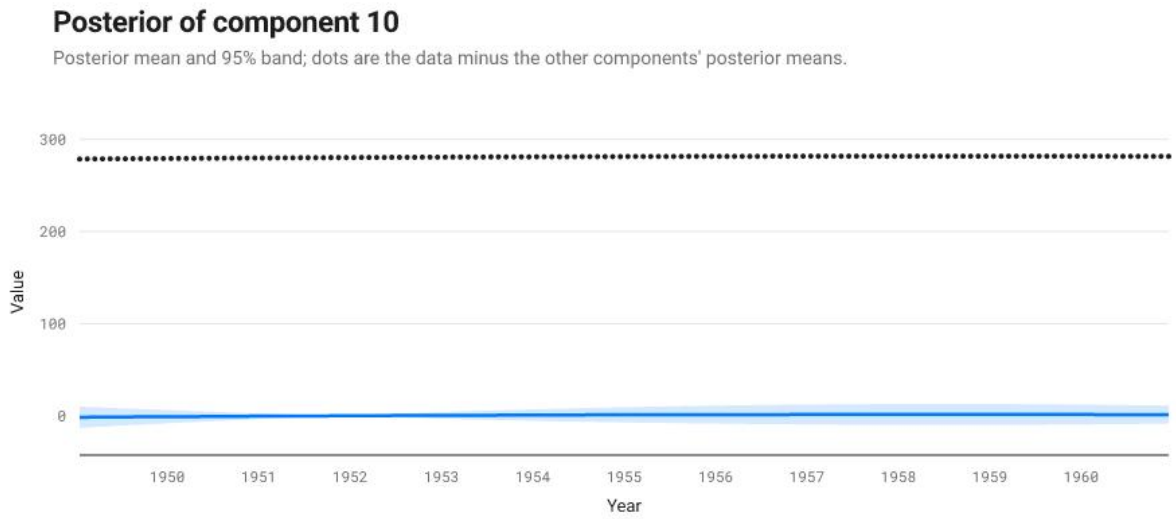
Component 9

Component 10: A quadratic polynomial

This component is a quadratic polynomial.

This component explains 1.2% of the data's variance on its own; with the components above it, 86.1%. Adding it changes the cross-validated MAE by -0.4%.

Extrapolation. This component is assumed to continue as a quadratic polynomial.



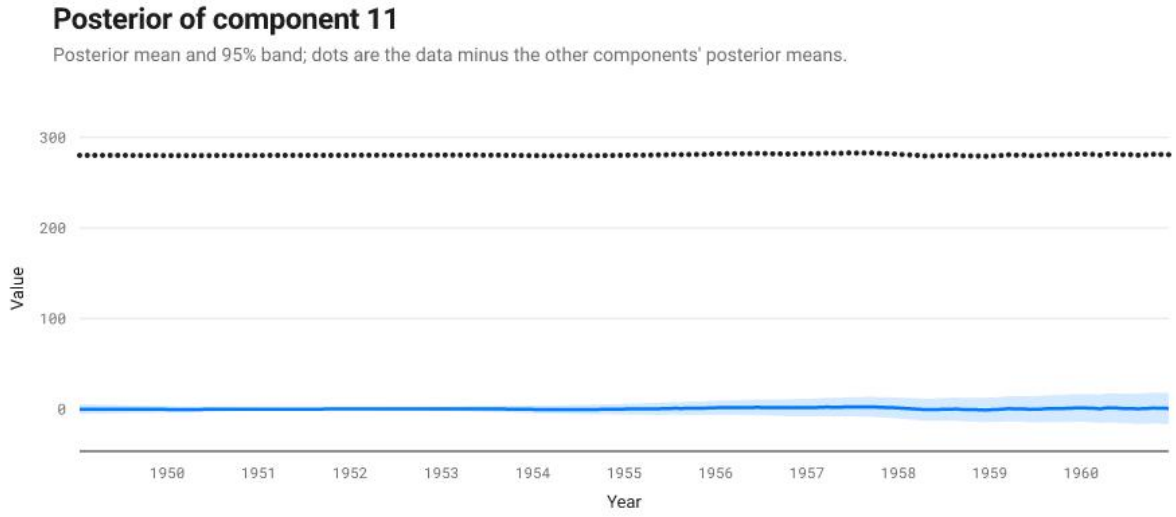
Component 10

Component 11: A rough function with marginal standard deviation increasing linearly away from Nov 1951

This component is a rough function with a typical lengthscale of 2.1 years. The marginal standard deviation of the function increases linearly away from Nov 1951. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.69).

This component explains 0.6% of the data's variance on its own; with the components above it, 86.2%. Adding it changes the cross-validated MAE by -0.2%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 2.1 years. The marginal standard deviation of the function is assumed to continue to increase linearly.



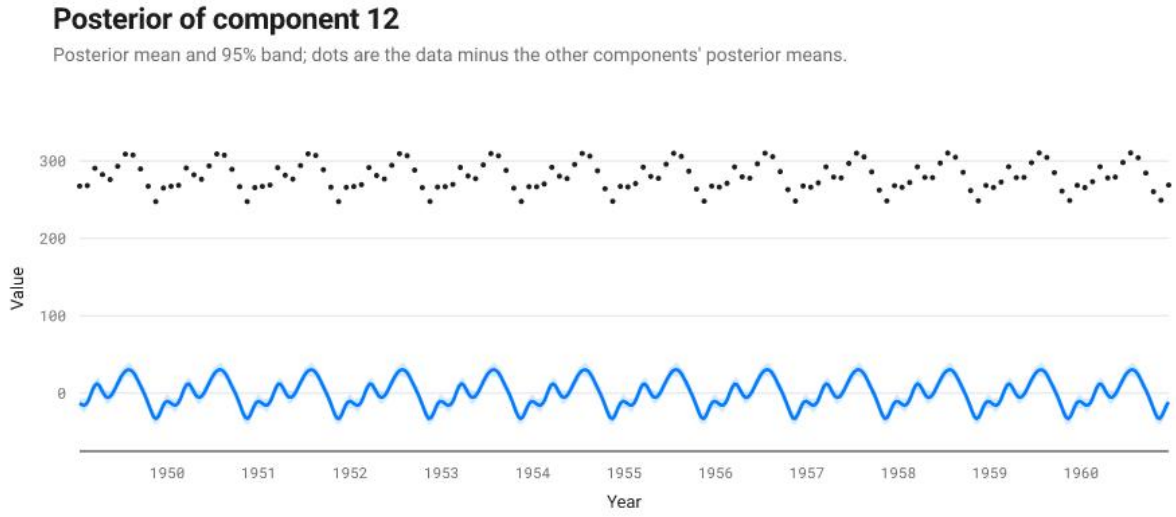
Component 11

Component 12: A periodic function with a period of 1.0 years

This component is periodic with a period of 1.0 years. The shape of this function within each period has a typical lengthscale of 4.0 weeks.

This component explains 6.4% of the data's variance on its own; with the components above it, 93.0%. Adding it changes the cross-validated MAE by -0.2%.

Extrapolation. This component is assumed to continue periodically with a period of 1.0 years.



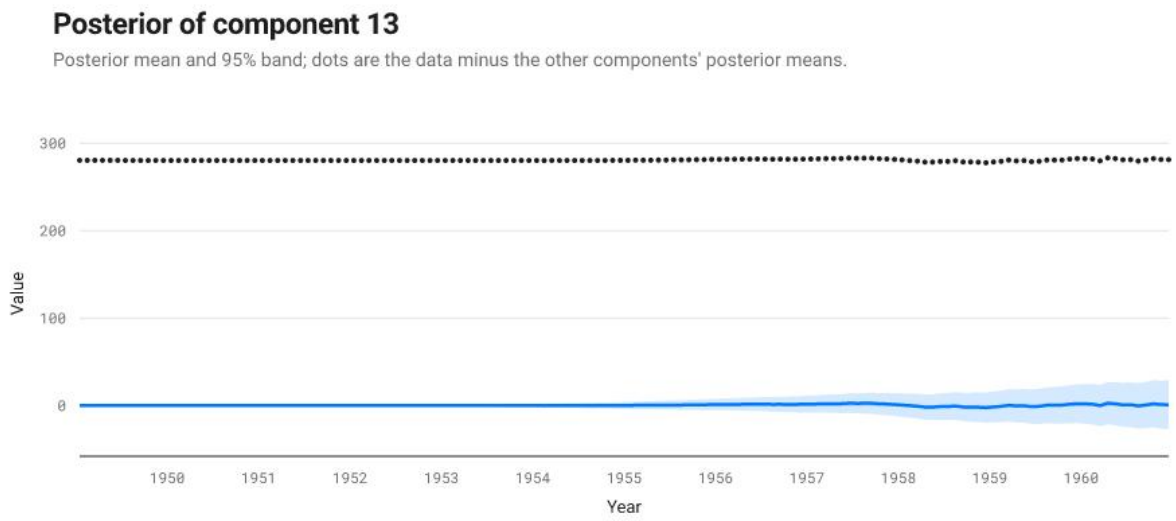
Component 12

Component 13: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 2.1 years. The marginal standard deviation of the function varies quadraticly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.69).

This component explains 0.4% of the data's variance on its own; with the components above it, 93.1%. Adding it changes the cross-validated MAE by -0.2%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 2.1 years. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



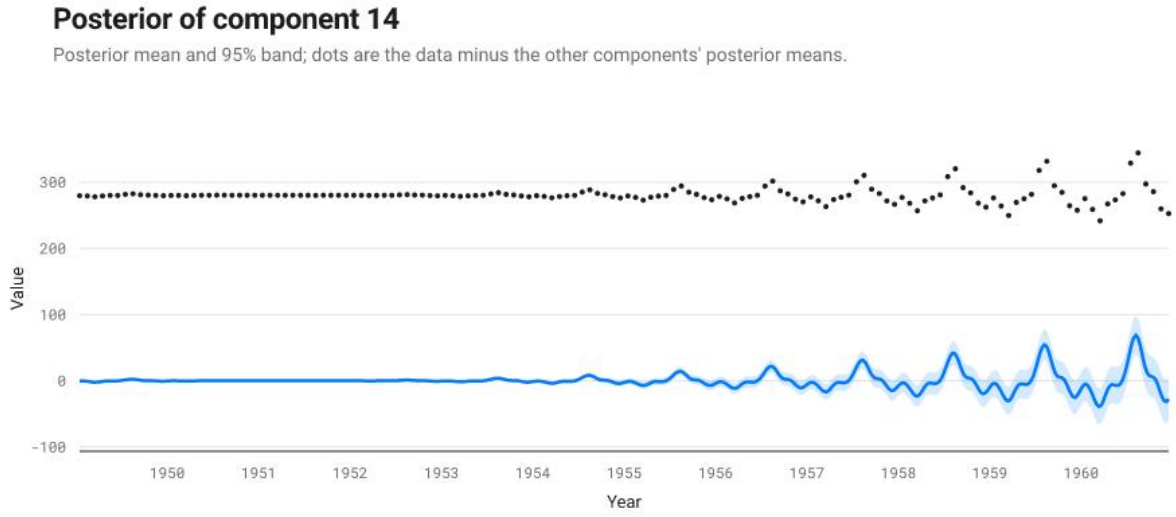
Component 13

Component 14: A periodic function with a period of 1.0 years but with quadraticly varying amplitude

This component is periodic with a period of 1.0 years but with varying amplitude. The amplitude of the function varies quadraticly. The shape of this function within each period has a typical lengthscale of 4.0 weeks.

This component explains 5.5% of the data's variance on its own; with the components above it, 96.8%. Adding it changes the cross-validated MAE by -0.1%.

Extrapolation. This component is assumed to continue periodically with a period of 1.0 years but with varying amplitude. The amplitude of the function is assumed to continue to vary quadraticly.



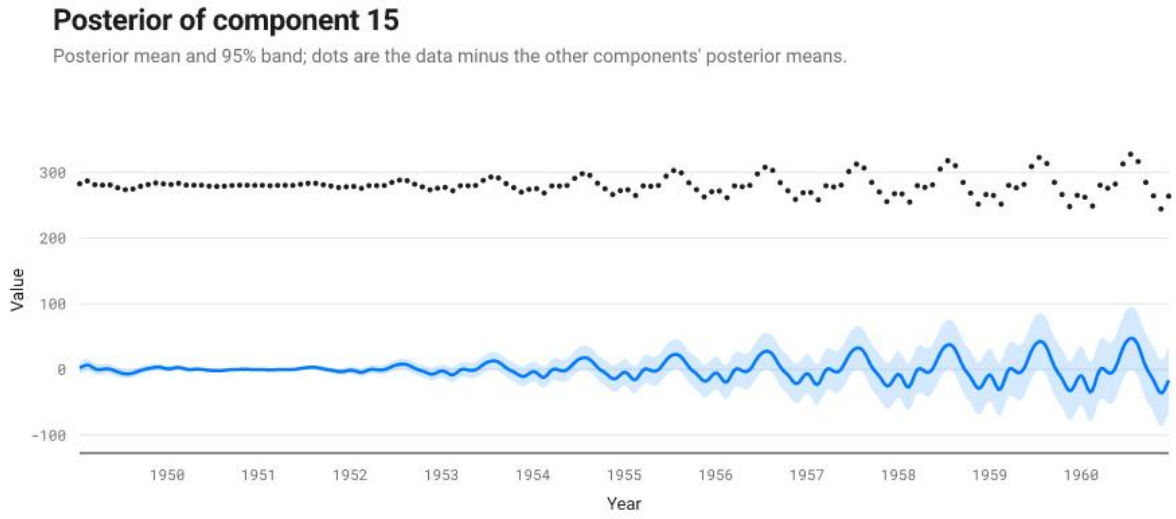
Component 14

Component 15: A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Dec 1950

This component is periodic with a period of 1.0 years but with varying amplitude. The amplitude of the function increases linearly away from Dec 1950. The shape of this function within each period has a typical lengthscale of 4.0 weeks.

This component explains 6.8% of the data's variance on its own; with the components above it, 99.2%. Adding it changes the cross-validated MAE by -0.1%.

Extrapolation. This component is assumed to continue periodically with a period of 1.0 years but with varying amplitude. The amplitude of the function is assumed to continue to increase linearly.



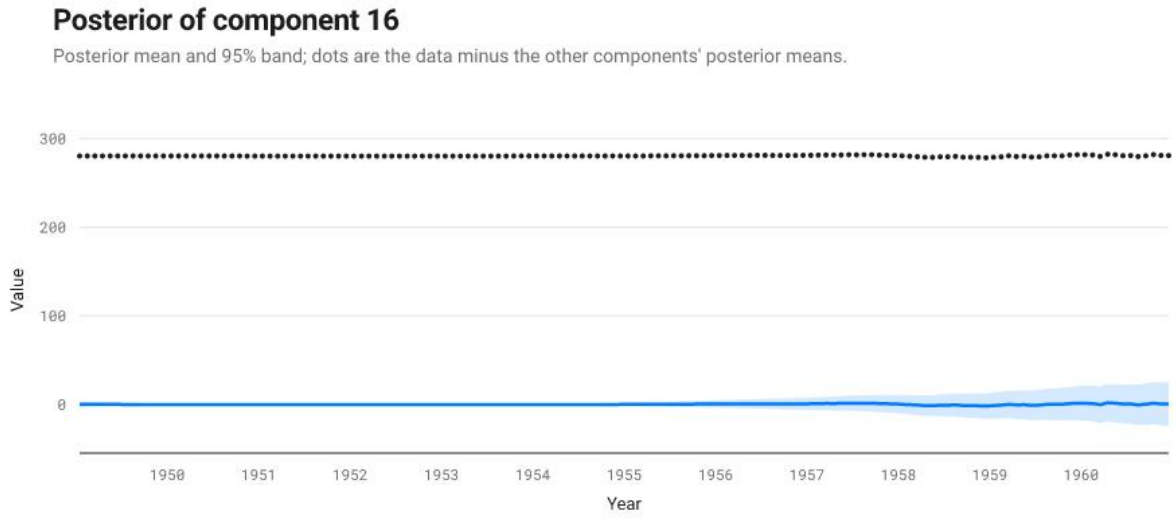
Component 15

Component 16: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 2.1 years. The marginal standard deviation of the function varies quadratically. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.69).

This component explains 0.2% of the data's variance on its own; with the components above it, 99.3%. Adding it changes the cross-validated MAE by -0.1%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 2.1 years. The marginal standard deviation of the function is assumed to continue to vary quadratically.



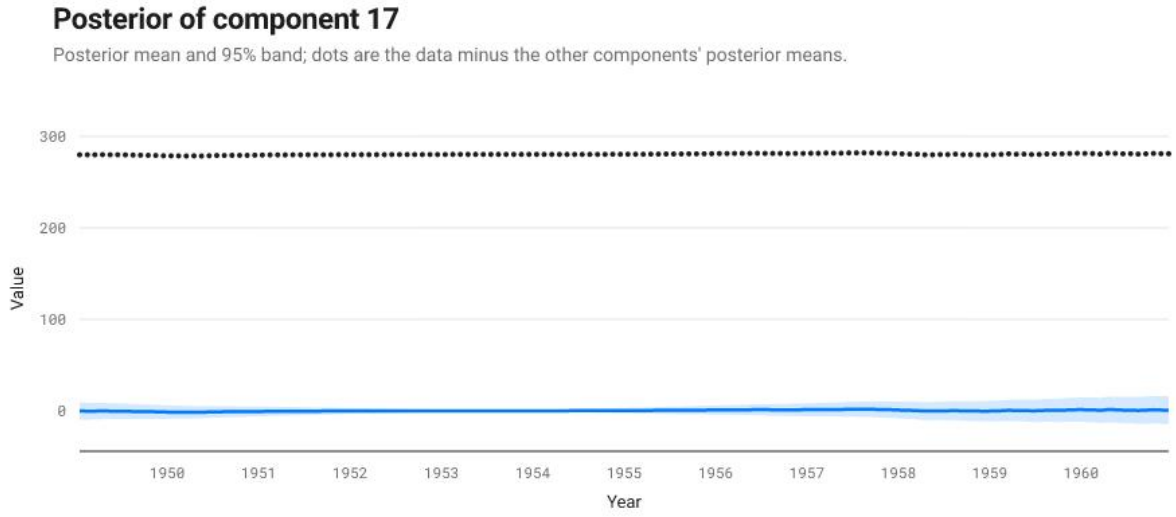
Component 16

Component 17: A rough function with marginal standard deviation increasing linearly away from Aug 1953

This component is a rough function with a typical lengthscale of 2.1 years. The marginal standard deviation of the function increases linearly away from Aug 1953. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.69).

This component explains 0.9% of the data's variance on its own; with the components above it, 99.3%. Adding it changes the cross-validated MAE by -0.1%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 2.1 years. The marginal standard deviation of the function is assumed to continue to increase linearly.



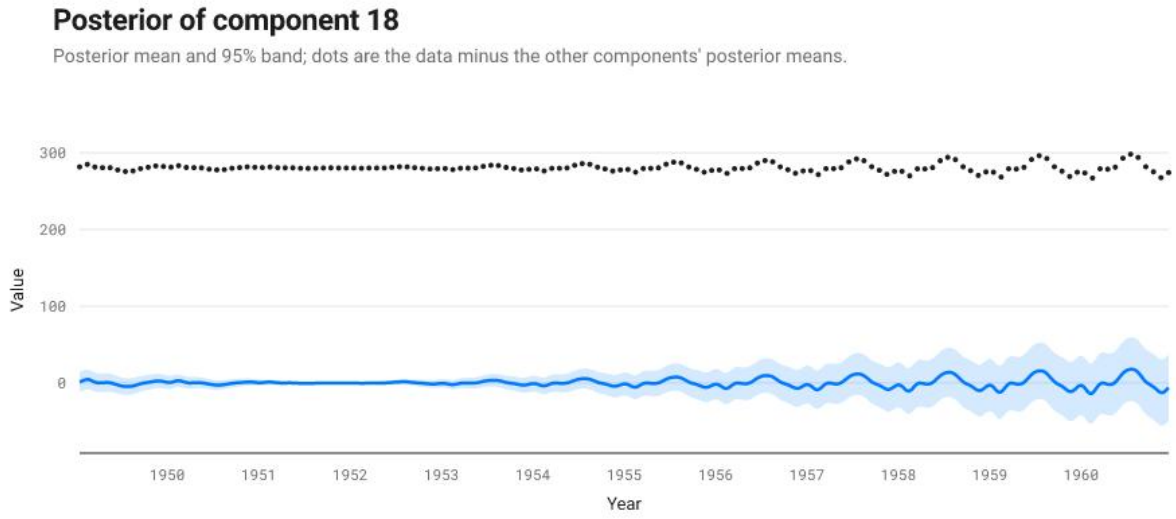
Component 17

Component 18: A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Nov 1951

This component is periodic with a period of 1.0 years but with varying amplitude. The amplitude of the function increases linearly away from Nov 1951. The shape of this function within each period has a typical lengthscale of 4.0 weeks.

This component explains 2.7% of the data's variance on its own; with the components above it, 99.5%. Adding it changes the cross-validated MAE by -0.1%.

Extrapolation. This component is assumed to continue periodically with a period of 1.0 years but with varying amplitude. The amplitude of the function is assumed to continue to increase linearly.



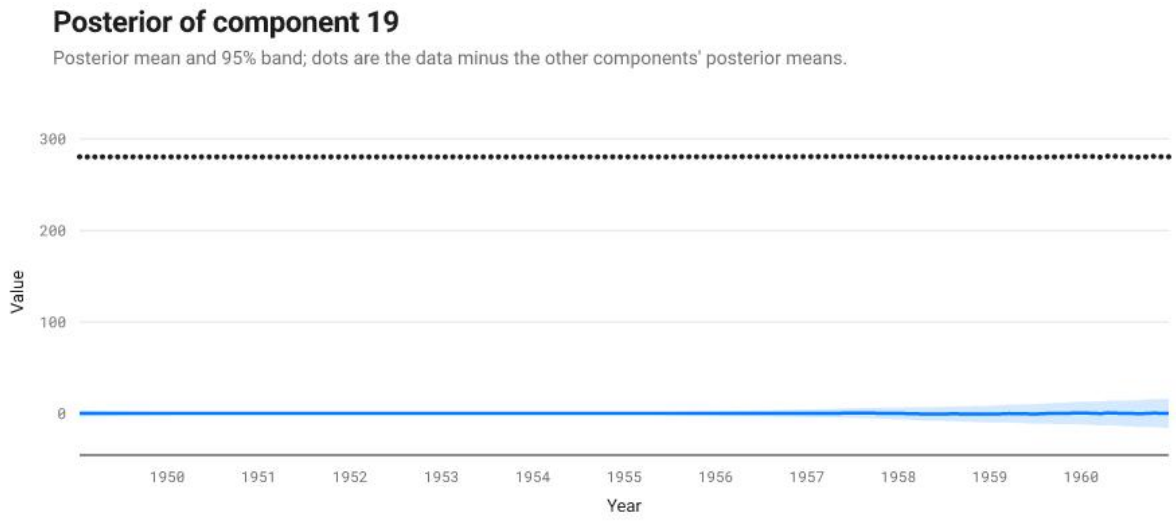
Component 18

Component 19: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 2.1 years. The marginal standard deviation of the function varies quadratically. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.69).

This component explains 0.0% of the data's variance on its own; with the components above it, 99.5%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 2.1 years. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



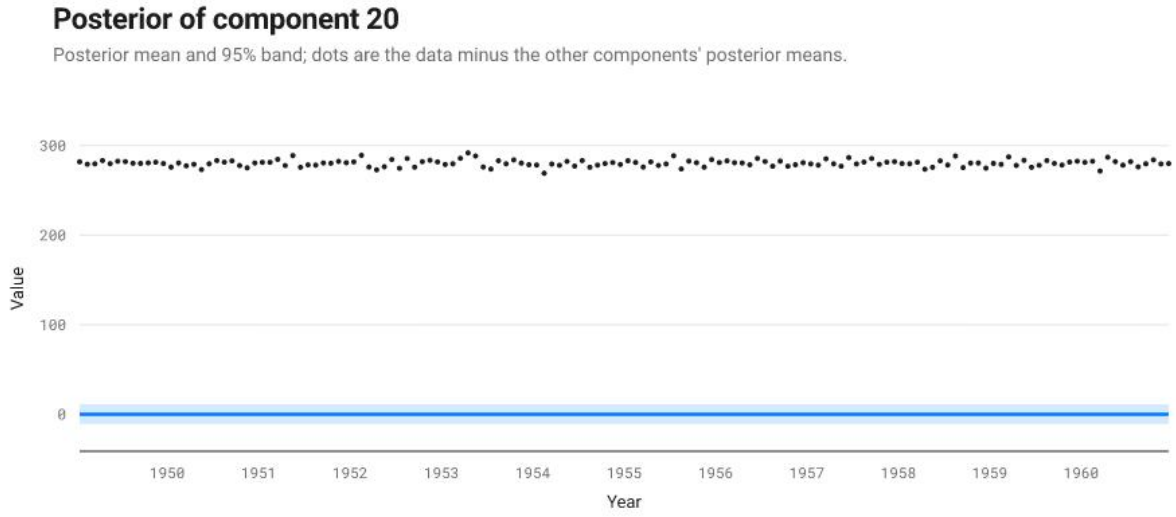
Component 19

Component 20: Uncorrelated noise

This component models uncorrelated noise.

This component explains 0.3% of the data's variance on its own; with the components above it, 99.7%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component assumes the uncorrelated noise will continue indefinitely.



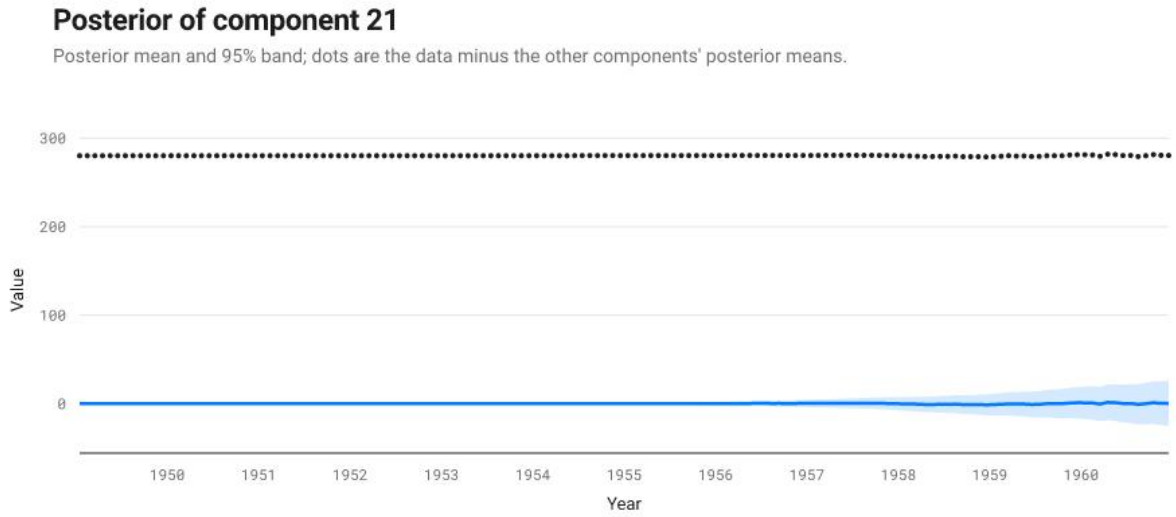
Component 20

Component 21: A rough function with cubicly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 2.1 years. The marginal standard deviation of the function varies cubicly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.69).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.7%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 2.1 years. The marginal standard deviation of the function is assumed to continue to vary cubicly.



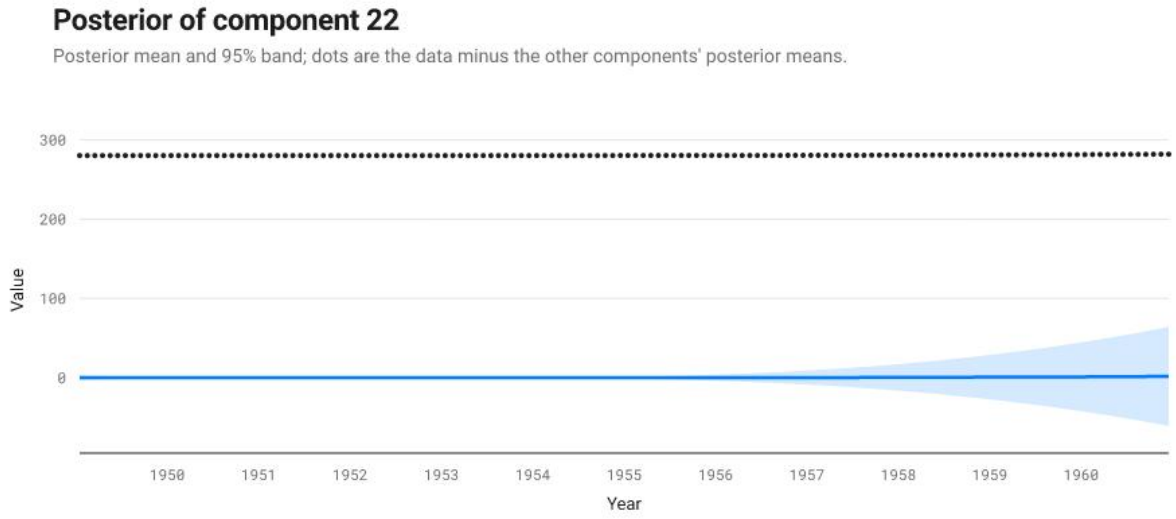
Component 21

Component 22: A cubic polynomial

This component is a cubic polynomial.

This component explains 0.6% of the data's variance on its own; with the components above it, 99.7%. Adding it changes the cross-validated MAE by 0.2%.

Extrapolation. This component is assumed to continue as a cubic polynomial.



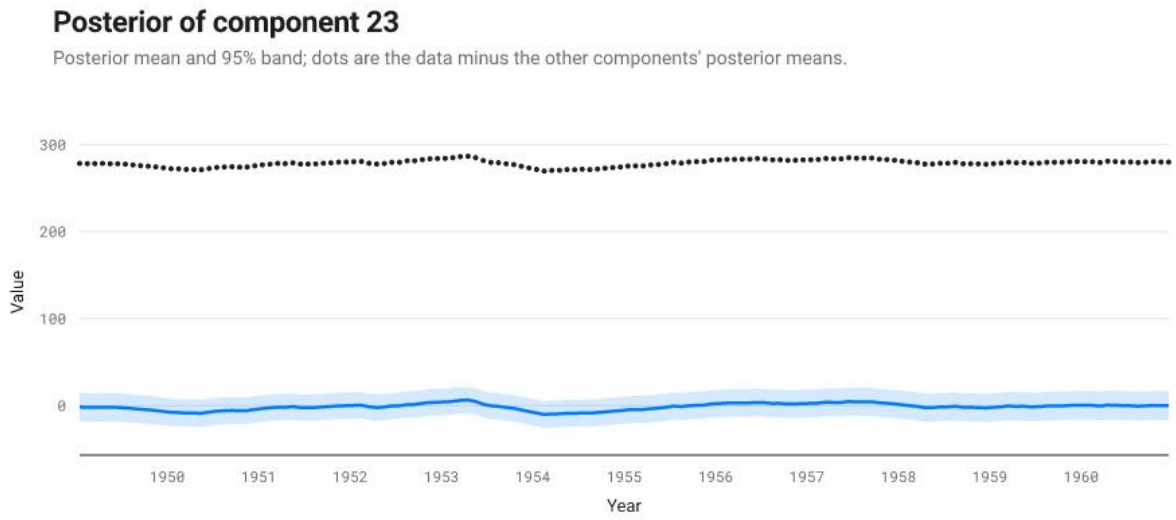
Component 22

Component 23: A rough function

This component is a rough function with a typical lengthscale of 2.1 years. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.69).

This component explains 2.5% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.6%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 2.1 years.



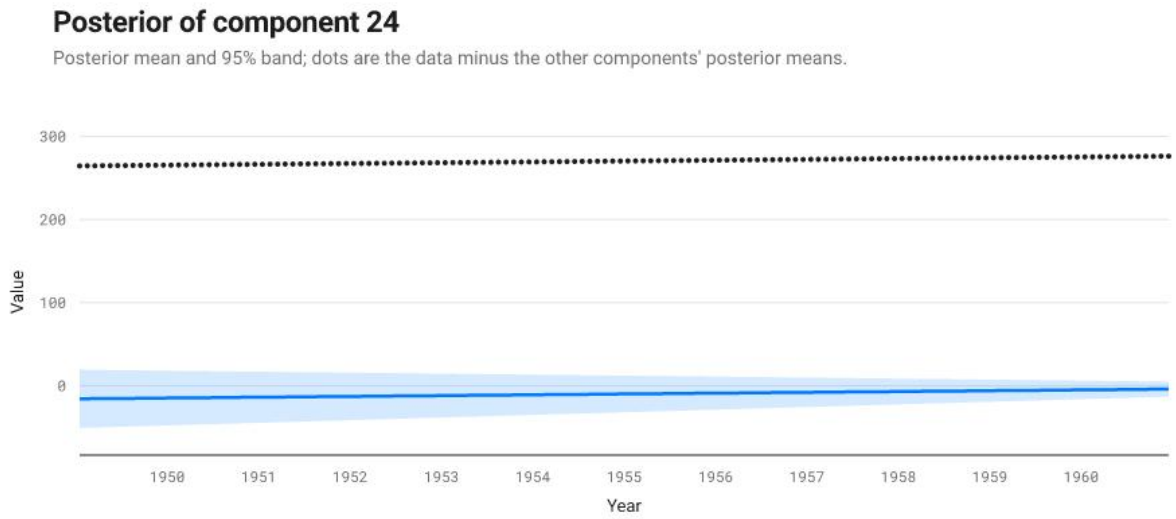
Component 23

Component 24: A linearly increasing function

This component is linearly increasing.

This component explains 5.2% of the data's variance on its own; with the components above it, 100.0%. Adding it changes the cross-validated MAE by 5.5%.

Extrapolation. This component is assumed to continue to increase linearly.



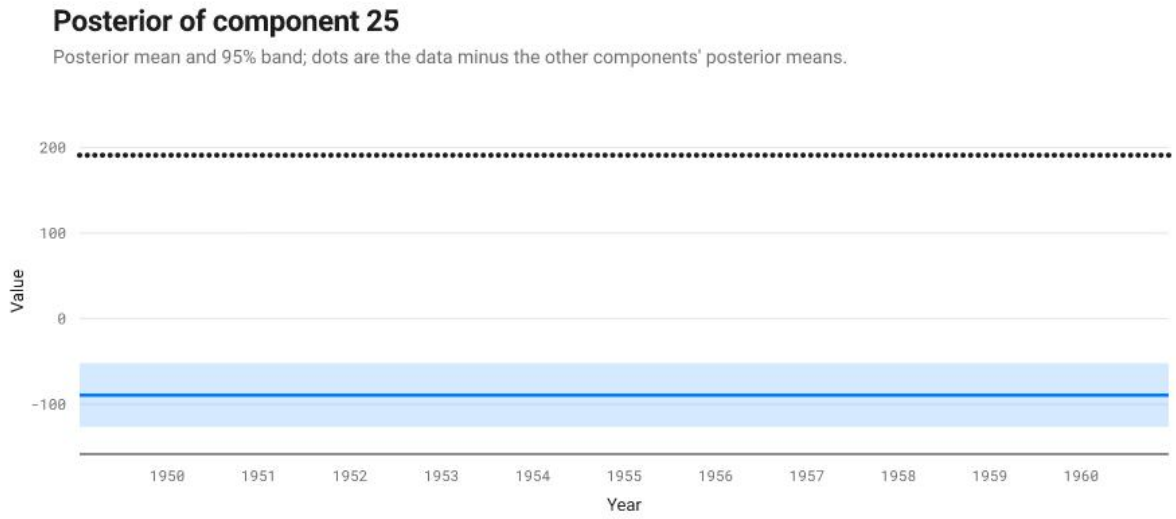
Component 24

Component 25: A constant

This component is constant.

This component explains 0.0% of the data's variance on its own; with the components above it, 100.0%. Adding it changes the cross-validated MAE by 46.5%.

Extrapolation. This component is assumed to stay constant.



Component 25

Model checking

Each statistic compares draws of a component's posterior with draws of its prior. A small p-value (bold below 0.05) points at a property of the data that the component's prior does not reproduce.

#	QQ max	QQ min	ACF min	ACF min lag	Pgram max	Pgram peak	MMD
1	0.920	0.228	0.980	0.988	0.480	0.974	**0.000**
2	0.700	0.490	0.974	0.968	0.796	0.998	**0.011**
3	0.878	0.366	0.980	0.994	0.590	0.998	**0.004**
4	0.842	0.820	0.988	0.970	0.986	0.980	0.395
5	0.990	0.712	0.936	0.984	0.732	0.904	0.601
6	0.870	0.956	0.916	0.946	0.954	0.980	**0.013**
7	0.924	0.886	0.986	0.974	0.944	0.980	0.083
8	0.918	0.928	0.974	0.960	0.996	0.938	0.125
9	0.880	0.910	0.974	0.990	0.922	0.990	0.647
10	0.976	0.934	0.972	0.960	0.978	0.996	0.392
11	0.906	0.930	0.978	0.950	0.976	0.988	0.897
12	0.606	0.494	0.694	0.966	0.504	0.466	**0.003**
13	0.960	0.972	0.996	0.966	0.976	0.930	0.431
14	0.678	0.424	0.926	0.990	0.842	0.570	**0.001**
15	0.896	0.912	0.932	0.962	0.870	0.730	**0.000**
16	0.956	0.974	0.956	0.962	0.988	0.994	0.234
17	0.974	0.962	0.982	0.996	0.992	0.980	0.241
18	0.988	0.956	0.992	0.930	0.910	0.886	**0.031**
19	0.990	0.982	0.996	1.000	0.990	0.990	0.692
20	0.982	0.898	0.966	0.970	0.986	0.986	0.233
21	0.982	0.972	0.992	0.984	0.974	0.966	0.788
22	0.896	0.986	0.988	0.944	0.862	0.984	**0.022**
23	0.724	0.890	0.894	0.912	0.934	0.968	0.051
24	0.480	0.478	0.976	0.992	0.818	0.996	0.239
25	0.080	0.080	0.948	0.742	0.170	0.982	**0.000**

QQ: quantile-quantile gaps. ACF: autocorrelation function. Pgram: periodogram. MMD: maximum mean discrepancy two-sample test.